Ex.NO:3 Breadth First Search

AIM:

To implement breath first search to traverse a graph level by level and find the shortest path (fewest edges)

PROGRAM:

```
from collections import deque
g = { 'A':['B','C'],
      'B':['D','E'],
      'C':['F'],
      'D':[],
      'E':['F'],
      'F':[]
}

def bfs(s):
    vis, q = [s], deque([s])
    while q:
        for x in g[q.popleft()]:
            if x not in vis:
                vis.append(x)
                q.append(x)
    return vis
print("BFS:", bfs('A'))
```

Output :

BFS : ['A', 'B', 'C', 'D', 'E', 'F']

RESULT:

The breadth first search program was implemented in python and executed successfully.